INSEAD The Business School for the World®

Process and Operations Management
Tutorial 1: Process Improvement

Stefanos Poulidis

PhD, Decision Sciences

stefanos.poulidis@insead.edu

My Research

Design and Governance of Human-AI systems

https://stefanospoulidis.github.io/

➢ Should AI Nudge You or Tell You What to Do?
(INSEAD Knowledge article)

➢ When does AI assistance undermine learning?
(Wharton Knowledge article)

stefanos.poulidis@insead.edu

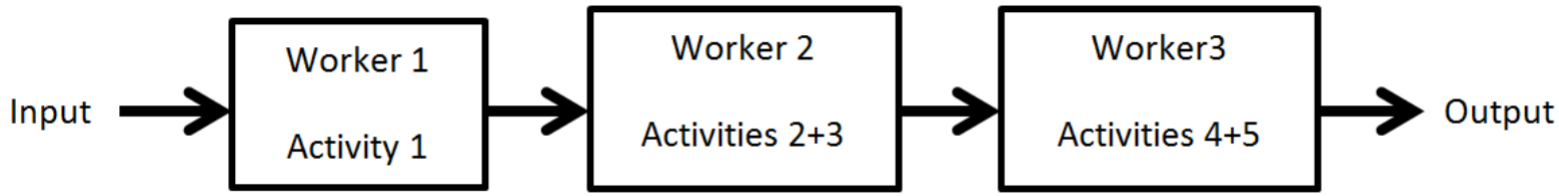
POM Tutorials

- **Session 1: Process Improvement**
- Session 2: Managing Variability
- Session 3: Managing Risks (The Newsvendor Model) I
- Session 4: Managing Risks (The Newsvendor Model) II
- Session 5: Q&A (final questions before the exam)

Note: I am not grading your assignments or final ☺

Process Improvement
Step 1: Draw a process flow diagram

Example 1 : Review Sheet 1, Problem 1. We have three resources:



Process Improvement
Step 2: Find the capacity of each resource

Capacity: the number of units of flow that a given resource can process per unit of time.

$$\text{Capacity of a resource} = \frac{1}{\text{average time spent per item by the resource}}$$

Example 2: If it takes 6 min (= 0.1 hours) to process an item, capacity is 1/6 units per min, or 10 units per hour.

Units should be consistent!

Process Improvement
Step 3: Find the bottleneck

- Case 1:

If in-flow = out-flow for all resources (i.e. there is no rework, no loss of yield, etc.), the bottleneck is **the resource with the lowest capacity**.

- Case 2:

If in-flow ≠ out-flow for at least one resource, the bottleneck is **the resource with the highest utilization**.

➢ The first one to reach 100% utilization as we increase inflow.

$$\text{Utilization of a resource}\,(\rho) = \frac{\text{Input rate to the resource}}{\text{Capacity of the resource}}$$

Process Improvement
Step 4: Find the system capacity

System capacity: System output per unit time when the bottleneck is fully (100%) utilized.

Instant Dolls Co.

Instant Dolls specializes in manufacturing dolls in a make to order basis. Their business process consists on the following activities, which must be conducted in the sequence described below:

Activity 1: Get the specifications of the doll from the customer, get the payment and submit the order to the production line. (8 minutes)

Activity 2: Assemble the main body of the doll (10 minutes)

Activity 3: Add hair and paint. (12 minutes)

Activity 4: Dress the doll (7 minutes)

Activity 5: Package the doll (3 minutes)

The whole process is conducted by three workers.

The assignment of tasks to workers is the following:

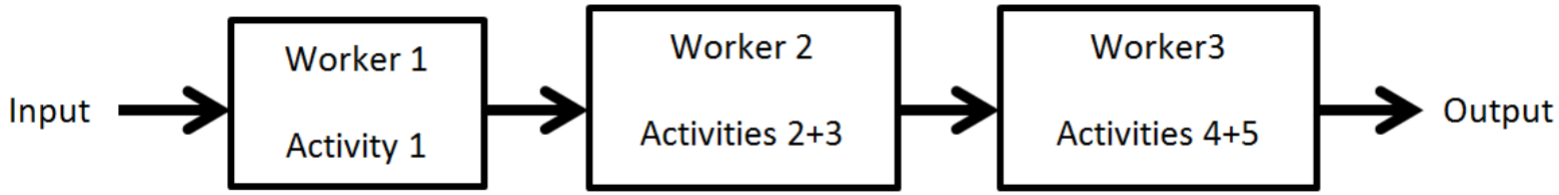
W1 does Activity 1.

W2 does Activities 2 and 3.

W3 does Activities 4 and 5.

Instant Dolls Co.

a) What is the bottleneck of the process?



Worker 1: 8 minutes

Worker 2: 10+12=22 minutes

Worker 3: 7+3=10 minutes

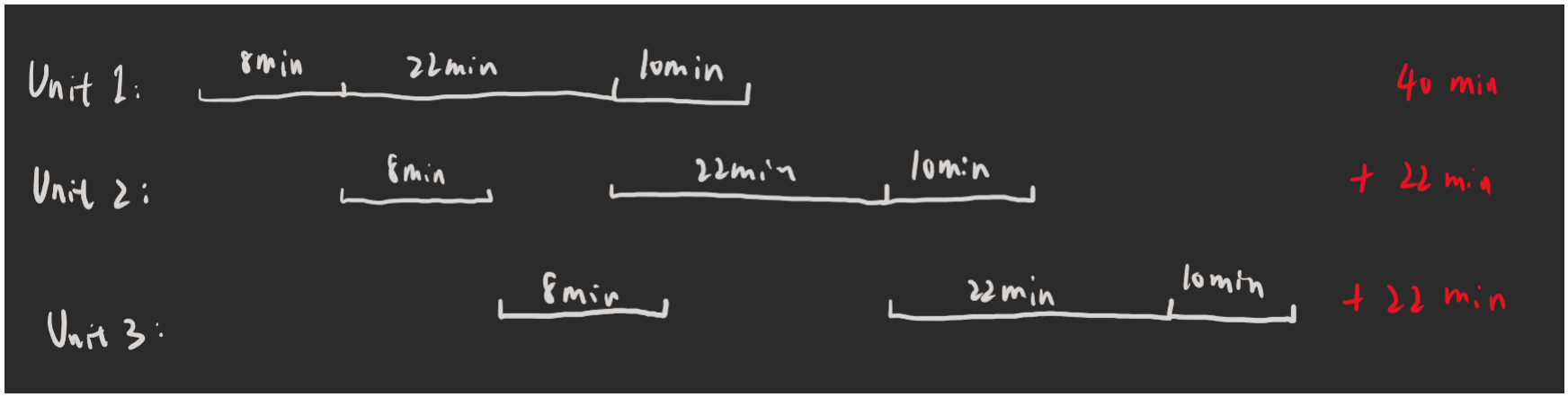
Instant Dolls Co.

b) How much time will it take to produce 250 units? (assume that the process starts with an empty production line)

40*250?

Worker 2 is the bottleneck and requires 22 mins.

22*250?



40+22*249=5518 mins

Instant Dolls Co.

c) What is the utilization of worker 3?

The input rate should be given to calculate the utilization rates. If not, we usually assume that plants should be operated with the bottleneck utilized 100%.

Bottleneck W2: 60/22=2.73 units/h

Capacity of W3: 60/10=6 units/h

Utilization rate of W3: 2.73/6=0.46

Instant Dolls Co.

d) What is the average labor utilization of the workers? Assume the process operates at its capacity and there are no empty system effects.

Flow rate=1/22 units/mins. Therefore:

W1 utilization= (1/22 units/min) / (1/8 units/min)= 8/22= 0.36

W2 utilization=(1/22 units/min) / (1/22 units/min) = 22/22 =1

W3 utilization=(1/22 units/min) / (1/10 units/min) =10/22= 0.45

Therefore, the average utilization is: (0.36+1+0.45)/3= 0.60.

Instant Dolls Co.

e) What are the direct labor costs for one doll? Assume a wage rate of $15 per hour.

As calculated in part b), the flow rate for the process is 1/22 units/min or 60/22 units/hour.

It follows that:

Cost of direct labor = total wages per hour/flow rate per hour

= (3 * $15 per h) / (60/22 units per hour)

= 16.5 per doll

Instant Dolls Co.

To increase the production rate, Instant Dolls is considering two alternatives:

Alternative I: To hire a new worker to help any of the workers without changing the tasks performed by each worker.

Alternative II: To redesign the assignment of tasks to workers. For this, the company is evaluating to re-assign Step 2 from W2 to W1.

f) Suppose that Alternative I is chosen. Where would you assign the new employee?

As seen with the calculations in part a), W2 has the highest utilization and so is the bottleneck. So I would assign the worker to W2.

Instant Dolls Co.

To increase the production rate, Instant Dolls is considering two alternatives:

Alternative I: To hire a new worker to help any of the workers without changing the tasks performed by each worker.

Alternative II: To redesign the assignment of tasks to workers. For this, the company is evaluating to re-assign Step 2 from W2 to W1.

g) What would be the direct labor of one doll under Alternative I? Assume there is sufficient demand (system operates at its capacity) and there are no empty system effects.

In this case, W2 which now consists two workers completes the work on each doll in 11 mins. It follows that the new flow rate is: 1/11 units/min or 60/11 units/hour (W2 is still the bottleneck: 8 mins for W1 and 10 mins for W3).

Using the same approach as in part e), we have:

Direct labor cost per doll = (4*15) / (60/11) = $11.

Instant Dolls Co.

To increase the production rate, Instant Dolls is considering two alternatives:

Alternative I: To hire a new worker to help any of the workers without changing the tasks performed by each worker.

Alternative II: To redesign the assignment of tasks to workers. For this, the company is evaluating to re-assign Step 2 from W2 to W1.

h) What would be the direct labor costs for Alternative II?

Worker 1: 8+10=18 minutes

Worker 2: 12 minutes

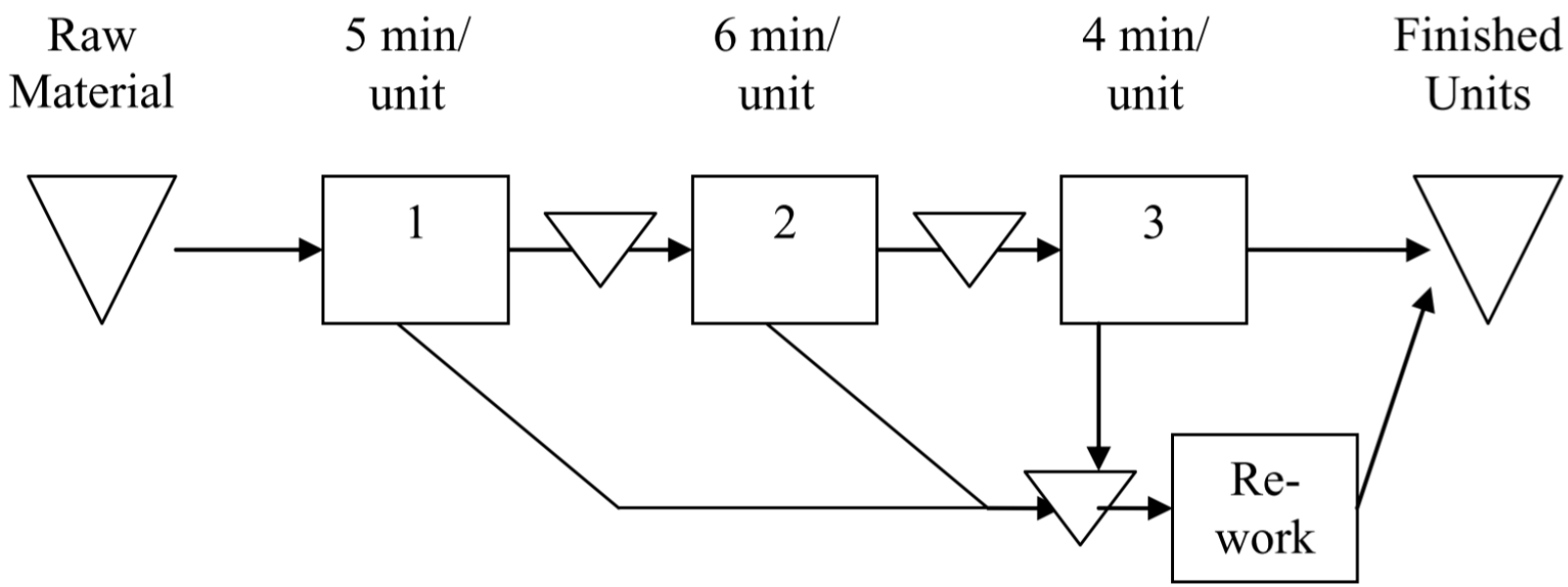
Worker 3: 7+3=10 minutes

It is easy to see that the new flow rate is: 1/18 units/min or 60/18 units/hour. Following the same approach as in part e), we have:

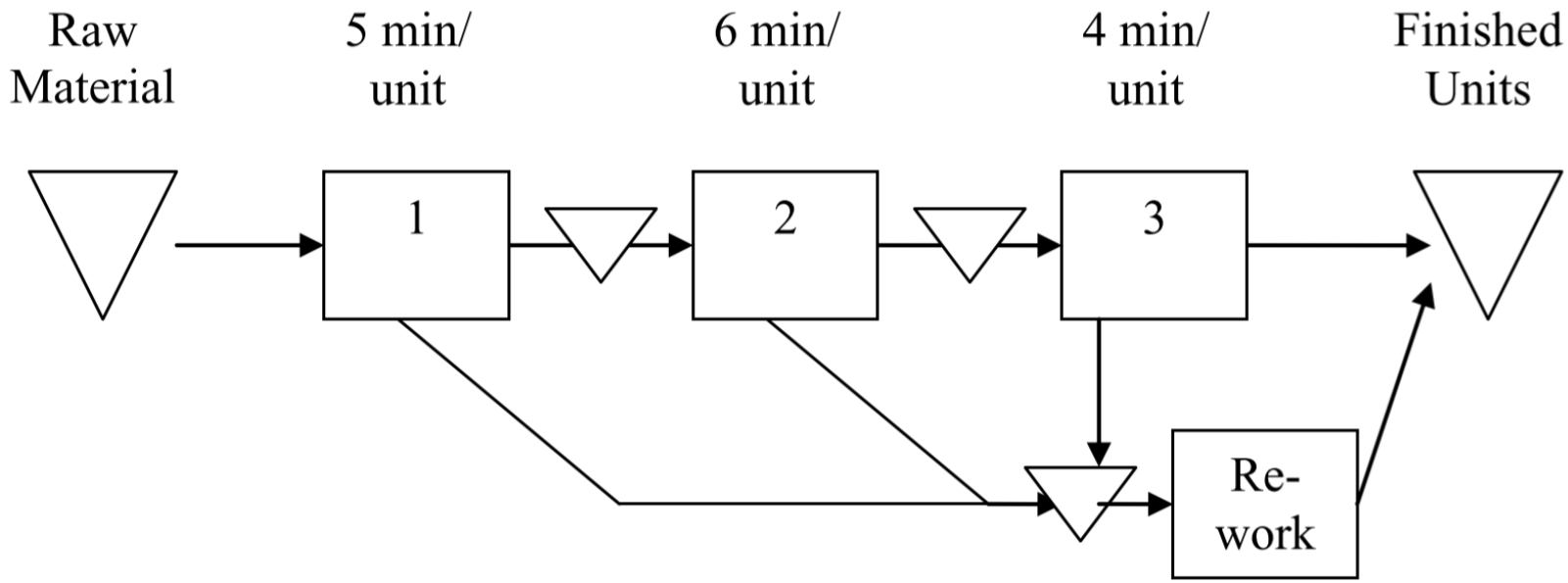
Direct labor cost per doll= (3*15) / (60/18)= $13.5.

Ceramics Production

Will Clintoon manages a glass ceramics production process that has three stages. Each stage operates at the rate that the operator at each stage can work. Squares represent process steps, and triangles represent the potential for work in process inventory.



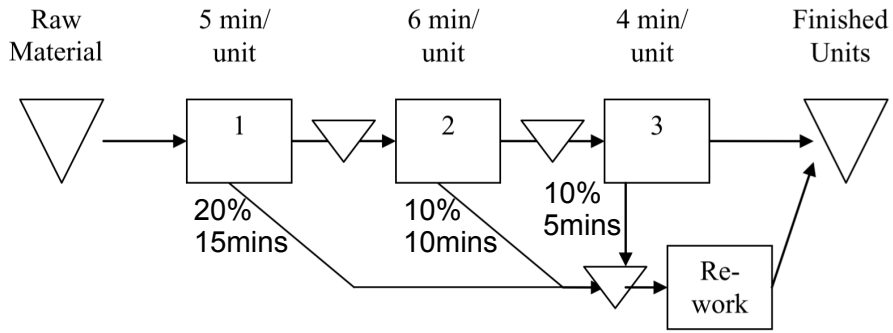
Ceramics Production



The process is experiencing serious quality problems related to management providing insufficient training for workers. Approximately 20% of the parts going through stage 1 are badly processed by the operator. Instead of scrapping the unit, it is sent to a highly skilled rework operator that can correct the mistake and finish the unit completely in 15 minutes.

The same problem occurs at stage 2 where 10% of the parts are poorly processed, and require 10 minutes of rework. Stage 3 also has a 10% defect rate, each defect requires 5 minutes of the rework operator time.

Ceramics Production



(i) What is the utilization of stage 2 if work is released into the system at a rate of 5 units per hour?

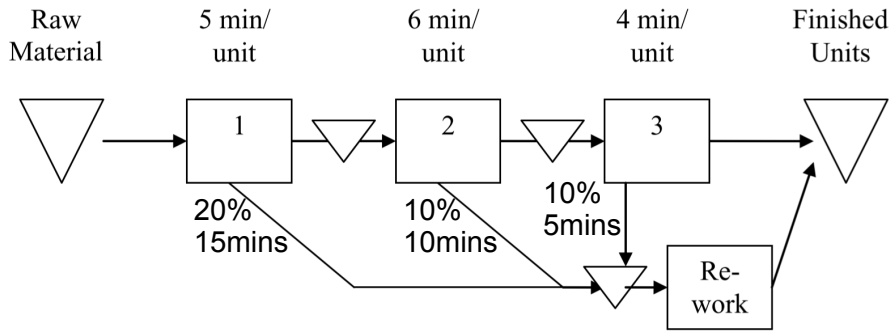
Capacity of stage 1 = 12 units per hour; Capacity of stage 2 = 10 units per hour; Capacity of stage 3 = 15 units per hour.

If the input rate to stage 1 is λ_1= 5 units/hour, then 20% of this input goes to rework operator. Thus, only 80% of this input reaches stage 2.

Stage 2 sees units coming at the rate of: λ_2 = λ_1 * 0.8 = 4 units/hour.

The capacity of stage 2 is 10 units per hour, so the utilization at stage 2 is ρ_2 = 4 unit per hour /10 units per hour = 0.4.

Ceramics Production



(ii) Where in the process is the bottleneck? Why?

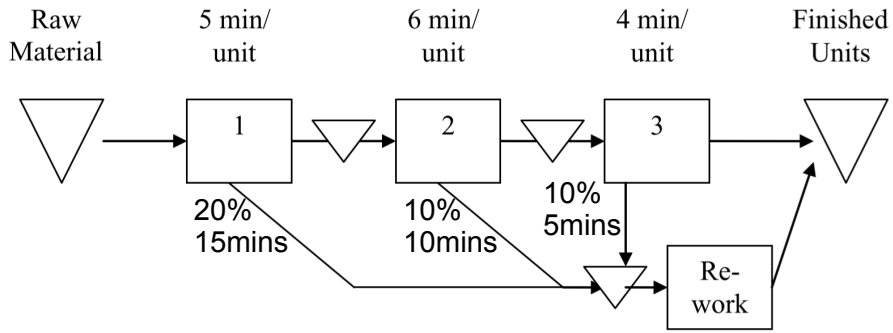
For stage 2, the process is just like as in (i): $\rho_2 = 0.4$.

ρ_1 = 5 units per hour/12 units per hour = 0.4167.

The arrival rate of work to stage 3 is $\lambda_3 = \lambda_2 * 0.9 = 4 * 0.9 = 3.6$ units per hour.
ρ_3 = 3.6 units per hour / 15 units per hour = 0.24.

For the rework stage, the task is trickier since three types of work might arrive that have three different processing times. We also note that the rework stage has different capacities for dealing with each type of flow.

Ceramics Production



(ii) Where in the process is the bottleneck? Why?

Rework operator can correct a defect coming from stage 1 at a rate of 4 units per hour. Similarly, he can correct defects coming from stage 2 at a rate of 6 units per hour and defects coming from stage 3 at a rate of 12 units per hour.

ρ_{R1} = 5 units per hour * 0.2 / 4 units per hour = 0.25

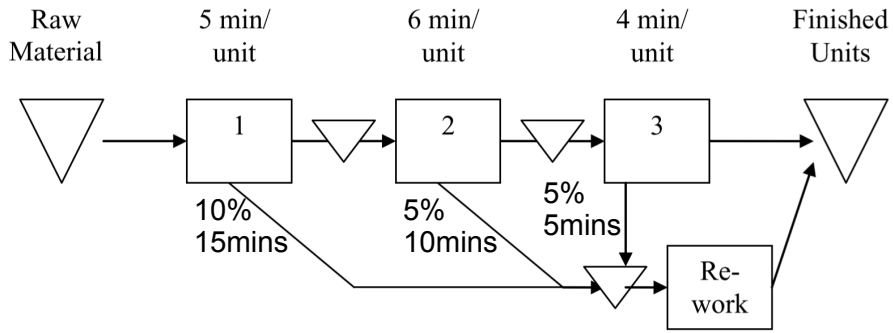
ρ_{R2} = 4 units per hour * 0.1 / 6 units per hour = 0.0666

ρ_{R3} = 3.6 units per hour * 0.1 / 12 units per hour = 0.03

$\rho_R = \rho_{R1} + \rho_{R2} + \rho_{R3} = 0.25 + 0.066 + 0.03 = 0.3466$

Stage 1 is the most heavily utilized and is currently the bottleneck. Stage 1 is busy 41.67% of the time.

Ceramics Production



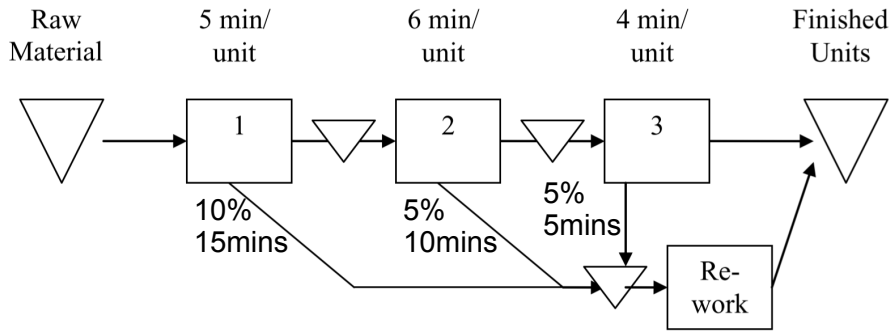
(iii) After a training program is implemented, the defect rate at stage 1 is reduced to 10% and the defect rates at stage 2 and stage 3 are both reduced to 5%. What is the maximum process capacity now?

Improving the defect rates changes the process flows:

Less flows to the rework area. Since it was not a bottleneck before, it will not become a bottleneck now.

There is an increase in flow to stages 2 and 3, so these might become bottlenecks now.

Ceramics Production



(iii) After a training program is implemented, the defect rate at stage 1 is reduced to 10% and the defect rates at stage 2 and stage 3 are both reduced to 5%. What is the maximum process capacity now?

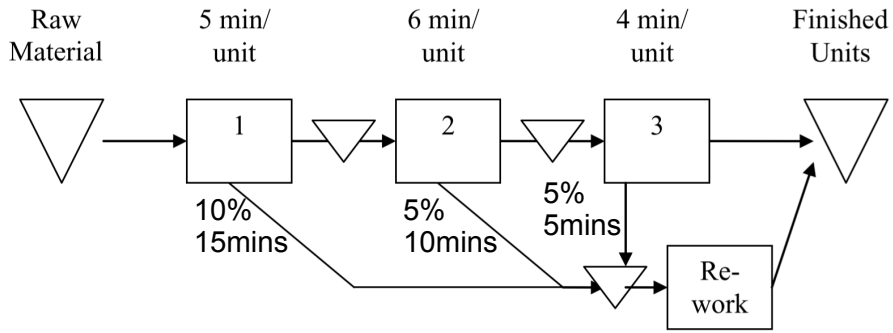
With a 10% defect rate at stage 1, stage 2 now sees an increased flow and utilization: $\lambda_2 = \lambda_1 * 0.9 = 4.5$ unit per hour

ρ_2 = 4.5 unit per hour / 10 units per hour = 0.45

Stage 3 also sees an increased utilization, but since it sees fewer jobs than stage 2 and work per unit is less, stage 3 will not be a bottleneck either.

Stage 1 is still seeing 5 units/hour, with capacity of processing 12 units per hour, for a 5/12=41.66% utilization. The quality improvements therefore changed the bottleneck from stage 1 to stage 2.

Ceramics Production



(iii) After a training program is implemented, the defect rate at stage 1 is reduced to 10% and the defect rates at stage 2 and stage 3 are both reduced to 5%. What is the maximum process capacity now?

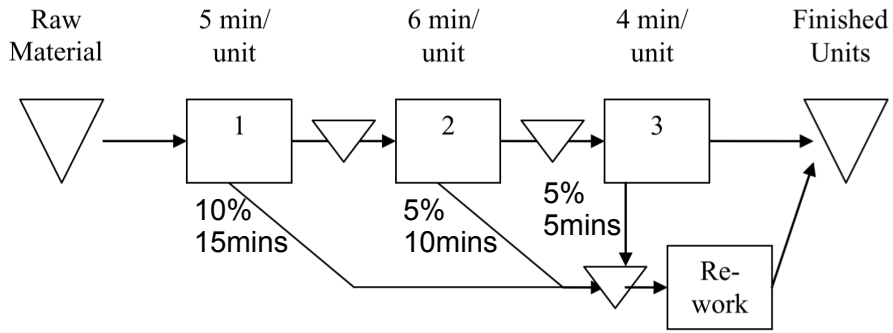
The theoretical maximum process capacity occurs when the bottleneck is 100% utilized. So the maximum arrival rate to stage 2 can be computed by seeing which rate corresponds to 100% utilization. In other words, we solve:

$\rho_2 = \lambda_{2,\,\mathrm{MAX}}$ /10 units per hour = 1.00

So the maximum input capacity at stage 2 is

$\lambda_{2,\,\mathrm{MAX}}$ = 10 units per hour

Ceramics Production



(iii) After a training program is implemented, the defect rate at stage 1 is reduced to 10% and the defect rates at stage 2 and stage 3 are both reduced to 5%. What is the maximum process capacity now?

To obtain a maximum capacity of 10 units per hour at stage 2, a higher rate of input to stage 1 is required, since 10% of the units processed at stage 1 go to Rework rather than stage 2.

Since $\lambda_2 = \lambda_1 * 0.9$, we have

$\lambda_{1,\,MAX} = \lambda_{2,\,MAX} / 0.9 = 10 / 0.9 = 11.11$ units per hour

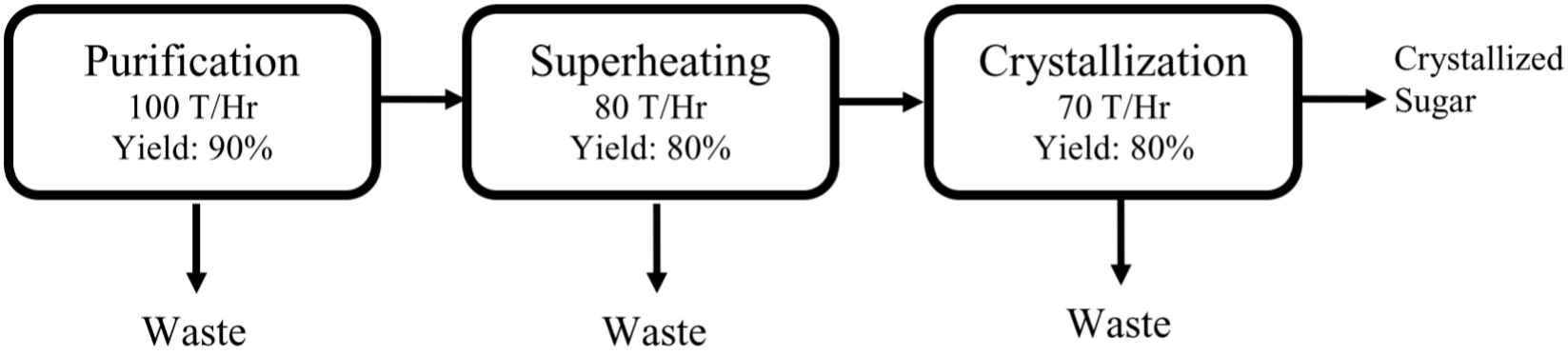
Of the 11.11 units per hour, 1.11 pass through rework instead of stage 2, but end up exiting the system eventually.

The overall maximum capacity is 11.11 units per hour.

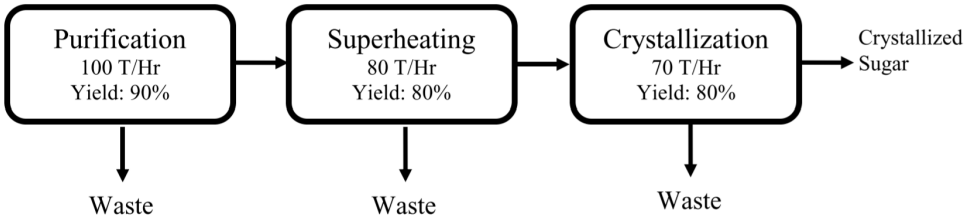
A Sweet Job

In an unexpected twist of events, you recently inherited a sugar plant from a distant relative. You are looking forward to apply your extensive knowledge of POM principles to this plant.

On your first visit to the sugar producing facility, you learn the following. Sugar production has basically three steps - Purification of Molasses followed by Superheating of the purified molasses and finally crystallization. Your facility has three extremely reliable continuous processing machines for the three process steps. To ensure the freshness of sugar, at no point in the process, can one build work in process inventories. The performance of the individual processing machines is described below.



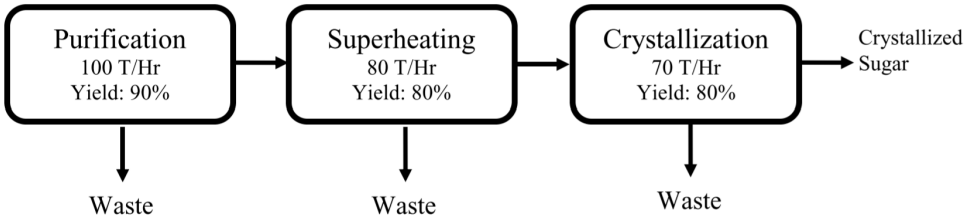
A Sweet Job



a) If 50 T/hr of molasses are introduced into the purification stage, what would be the utilization of each of the stages? What stage is the bottleneck? What is the capacity of this plant (in Tons of crystallized sugar per hour)?

Stage	*Purification*		*Superheating*		*Crystallization*	
	In	*Out*	*In*	*Out*	*In*	*Out*
Capacity	*100*		*80*		*70*	
Yield	*90%*		*80%*		*80%*	
A.	*50*	*45*	*45*	*36*	*36*	*28.8*
Utilization	***0.5***		***0.5625***		***0.514286***	

A Sweet Job



a) If 50 T/hr of molasses are introduced into the purification stage, what would be the utilization of each of the stages? What stage is the bottleneck? What is the capacity of this plant (in Tons of crystallized sugar per hour)?

To compute the capacity of the plant, the bottleneck would be fully utilized.

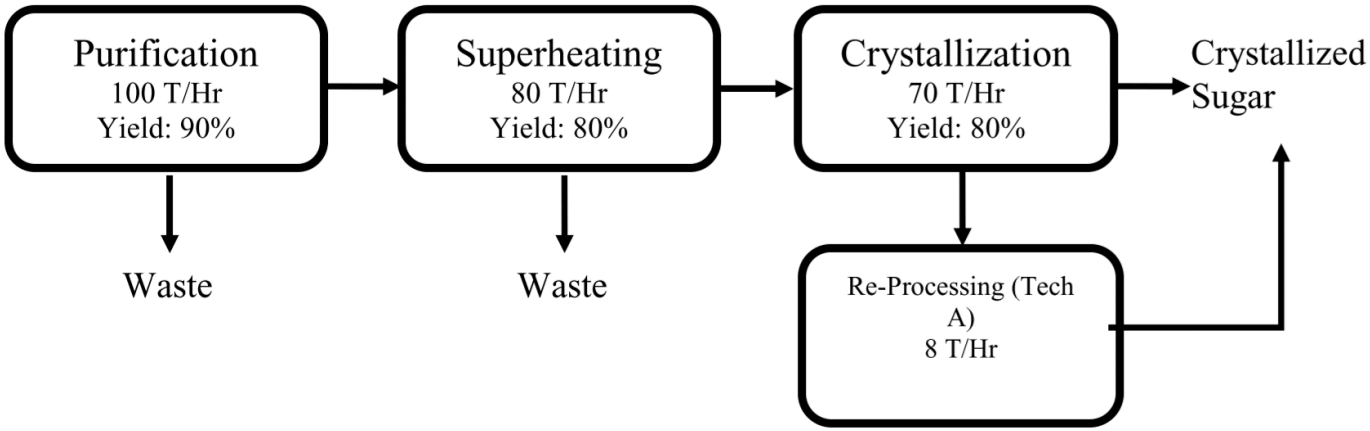
So the input to the superheating stage must be 80 T/hr.

Because superheating produces only 80% good output, then crystallization receives 80T/hr x 0.8 = 64 T/hr input. This stage also has 20% waste, so the final production would be 64 T/hr x 0.8 = 51.2 T/hr, which is the production line capacity.

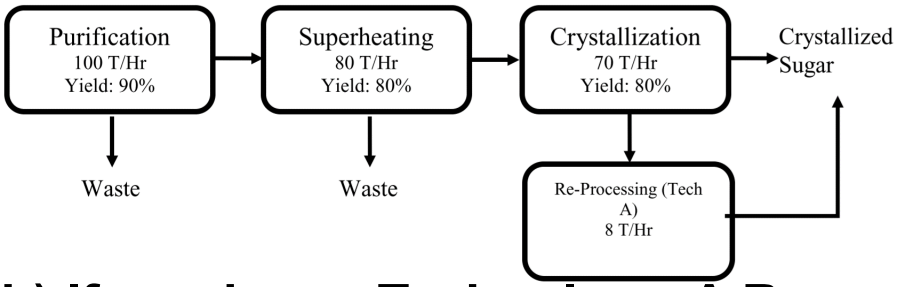
A Sweet Job

After analyzing the plant, you feel a little frustrated by the high yield losses. You contact your classmate from INSEAD, who is now the leading process improvement consultant for the sugar industry. He points out that there are a couple of new technologies available that can re-process the losses from the crystallization stage. This could increase the total capacity of the plant.

Technology A can take the losses from the crystallization stage and convert all the waste to crystallized sugar (the final product). (For efficient operation of the re-processing stage, all the waste that comes from the crystallization stage must be introduced into the Re-processing stage.) The modified Process diagram is illustrated below.



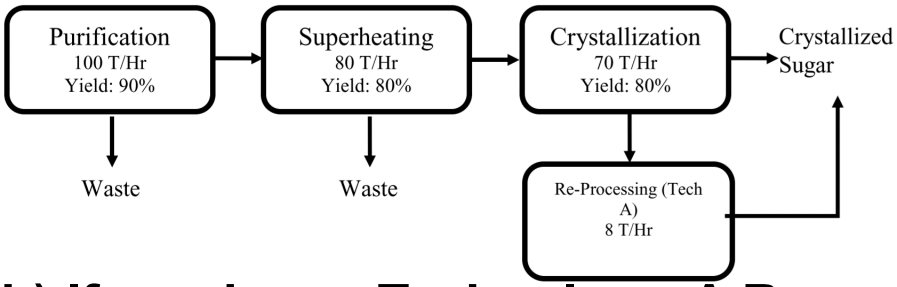
A Sweet Job



b) If you buy a Technology-A Re-processing machine with a capacity of 8 T/Hr, what is the new capacity of the plant? What is the minimum capacity of the new machine that would not change the bottleneck from the one you found in part a) above?

Stage	*Purification*		*Superheating*		*Crystallization*		*Re-Processing*	
	In	*Out*	*In*	*Out*	*In*	*Out*	*In*	*Out*
Capacity	*100*		*80*		*70*		*8*	
Yield	*90%*		*80%*		*80%*		*100%*	
A.	*50*	*45*	*45*	*36*	*36*	*28.8*	*7.2*	*7.2*
Utilization	*0.5*		*0.5625*		*0.514286*		*0.9*	

A Sweet Job

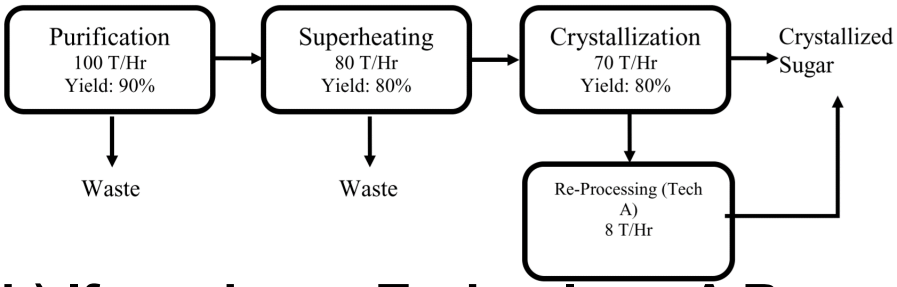


b) If you buy a Technology-A Re-processing machine with a capacity of 8 T/Hr, what is the new capacity of the plant? What is the minimum capacity of the new machine that would not change the bottleneck from the one you found in part a) above?

To get a 100% utilization of the re-processing stage, we need to introduce 40 T/hr in the crystallization and get a 40 T/hr × 0.2 = 8 T/hr waste to keep the bottleneck busy.

Crystallization then produces 32 T/hr and Re-processing finishes the 8 T/hr and so the capacity of the process is 40 T/hr.

A Sweet Job



b) If you buy a Technology-A Re-processing machine with a capacity of 8 T/Hr, what is the new capacity of the plant? What is the minimum capacity of the new machine that would not change the bottleneck from the one you found in part a) above?

To ensure there is no new bottleneck, the re-processing stage should not be utilized any more than 56.25% for 50 T of input, which is the utilization of superheating stage for 50 T/hr input.

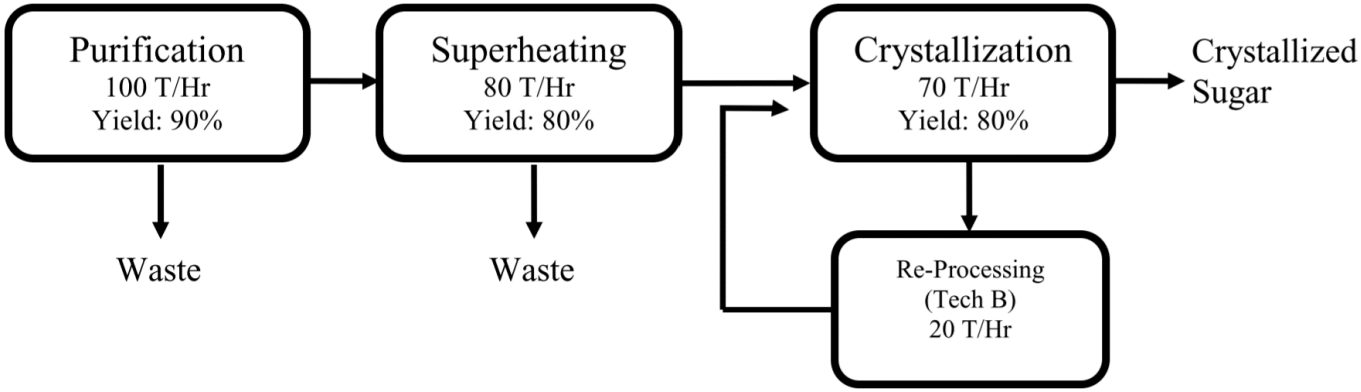
As calculated in the table, for this amount of input into the production line, the amount of input that gets to the re-processing stage is 7.2 T/hr.

So the capacity must be such that: 7.2 / Capacity = 0.5625 or capacity = 7.2/(0.5625) = 12.8 T/hr.

Technology A is NOT good unless it has a capacity higher than 12.8 T/hr.

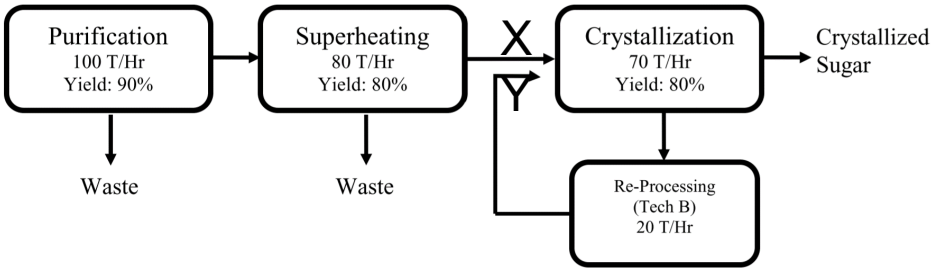
A Sweet Job

Technology B on the other hand takes the losses from the crystallization stage and converts them to super-heated liquid, which must be again re-introduced into the crystallization stage. The process diagram is illustrated below



c) If you buy a Technology-B Re-processing machine with a capacity of 20 T/Hr, what is the new capacity of the plant? (Like Technology A, efficient operation of Technology B requires that all the waste from the crystallization stage be introduced into the reprocessing stage)

A Sweet Job



c) If you buy a Technology-B Re-processing machine with a capacity of 20 T/Hr, what is the new capacity of the plant? (Like Technology A, efficient operation of Technology B requires that all the waste from the crystallization stage be introduced into the reprocessing stage)

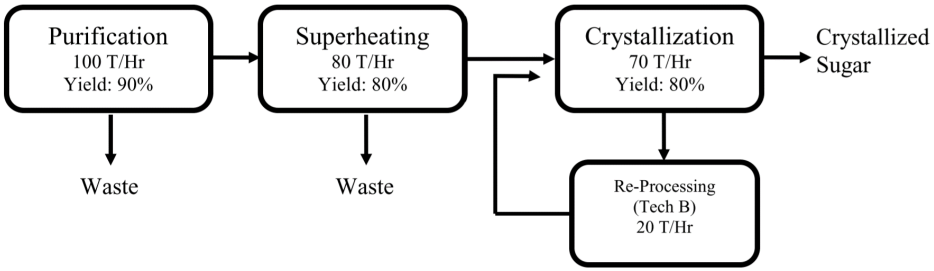
The only hard thing in this case is to note that some of the output gets recycled.

Say at the crystallization stage, X units come from superheating and Y units from the reprocessing stage.

Then, we can write a flow equation as Y=0.2(X+Y). Or Y=X/4.

Or total flow into Crystallization is 5/4 times the output of the superheating stage.

A Sweet Job

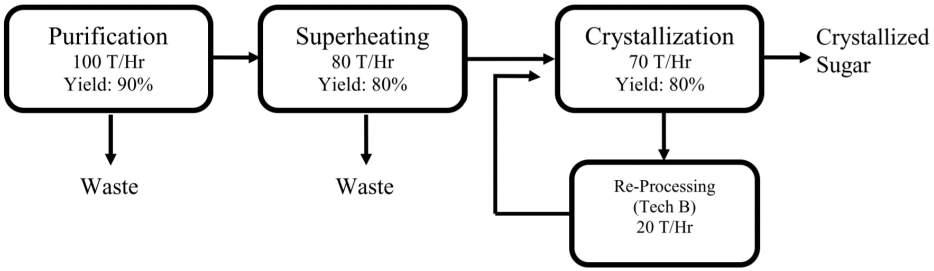


Total flow into Crystallization is 5/4 times the output of the superheating stage.

c) If you buy a Technology-B Re-processing machine with a capacity of 20 T/Hr, what is the new capacity of the plant? (Like Technology A, efficient operation of Technology B requires that all the waste from the crystallization stage be introduced into the reprocessing stage)

Stage	*Purification*		*Superheating*		*Crystallization*		*Re-Processing*	
	In	*Out*	*In*	*Out*	*In*	*Out*	*In*	*Out*
Capacity	*100*		*80*		*70*		*20*	
Yield	*90%*		*80%*		*80%*		*100%*	
A.	*50*	*45*	*45*	*36*	*45*	*36*	*9*	*9*
Utilization	*0.50*		*0.5625*		*0.6429*		*0.45*	

A Sweet Job



c) If you buy a Technology-B Re-processing machine with a capacity of 20 T/Hr, what is the new capacity of the plant? (Like Technology A, efficient operation of Technology B requires that all the waste from the crystallization stage be introduced into the reprocessing stage)

As for the previous case, to compute the capacity of the process the bottleneck would need to be fully utilized.

(70/45) * 36 = 56 T/Hr.

The calculation then shows that the capacity of this system is 56 T/Hr.

Sport Obermeyer

Sport Obermeyer manufactures parkas using an assembly line. It owns two factories, one in Mexico and the other in China. The workers in the Chinese factory are multi-skilled. The workers in Mexico are relatively less skilled and more specialized. Sport Obermeyer uses an assembly line with 20 workers in China and one with 40 workers in Mexico. The capacities are the same in the two assembly lines. Based on this information you can conclude that relative to the factory in China (explain to get credit):

a. The Mexican factory makes more parkas per hour but takes less time to make parkas.

b. The Mexican factory makes more parkas per hour but takes longer to make parkas.

c. The Mexican factory makes the same number of parkas per hour but takes longer to make parkas.

d. The Mexican factory makes the same number of parkas per hour but takes the same time to make parkas.

e. None of the above

Sport Obermeyer

We use Little's Law to solve the problem: the long-term average number L of customers in a stationary system is equal to the long-term average effective arrival rate λ multiplied by the average time W that a customer spends in the system.

The capacities in the two factories are the same. Therefore the throughput is the same, assuming the factories are working at full capacity.

Inventory is the work in progress. There are 40 parkas on the assembly line in Mexico at any given point (we are assuming no idle workers). There are just 20 parkas on the assembly line in China at any given point.

Then it must be that the flow time in China is lower than the one in Mexico.

The idea is that more unskilled workers (more steps in the manufacturing) lead to more inventory lying around. More work in progress in one factory, but same capacity translates into longer flow time in that factory.

Answer: c